

Variational Monte Carlo for the Bose–Hubbard Model

Code Documentation

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1 Overview

This document describes the implementation of a variational Monte Carlo (VMC) solver for the Bose–Hubbard (BH) model on a 1D chain. The code estimates the ground-state energy E_0 by stochastically sampling a parameterized trial wavefunction Ψ_T using the Metropolis–Hastings algorithm.

2 Model Hamiltonian

The Bose–Hubbard Hamiltonian is

$$\hat{H}_{\text{BH}} = -t \sum_{\langle i,j \rangle} \left(\hat{b}_i^\dagger \hat{b}_j + \text{h.c.} \right) + \frac{U}{2} \sum_i \hat{n}_i (\hat{n}_i - 1) - \mu \sum_i \hat{n}_i, \quad (1)$$

where t is the hopping amplitude, U is the on-site interaction strength, μ is the chemical potential, $\hat{b}_i^{(\dagger)}$ are the bosonic annihilation (creation) operators, and $\hat{n}_i = \hat{b}_i^\dagger \hat{b}_i$ is the number operator on site i . For our purposes, we ignore the chemical potential since move proposals never

3 Code Parameters

Below are all the parameters which you can control when running this VMC code. Most are self-explanatory, but I will take some time to explain the not-so-obvious parameters; specifically, I will discuss the parameters under "Optimization MC" and "Final MC." We distinguish between these two types of Monte Carlo as we perform shorter Monte Carlo simulations (Optimization MC) to measure the energy and energy gradient for gradient descent to obtain the optimal parameters, then use these optimized parameters in a longer Monte Carlo simulation (Final MC) to obtain precise ground state energy measurements.

num-walkers is the number of separate system configurations/Markov chains that we perform Monte Carlo moves on. **num-MC-steps** is the number of Monte Carlo moves the code attempts to make on the walkers/configurations. **num-equil-steps** is the number of Monte Carlo moves the code attempts to make before making measurements to ensure the system is equilibrated. **block-size** is the size of each block of averaged measurements in the Monte Carlo simulations.

Table 1: Command-line options for the BH-VMC code.

Flag	Type	Default
<i>Physical parameters</i>		
-L, --length	Int64	required
-N, --particle-number	Int64	required
-U, --interaction	Float64	required
-t, --t	Float64	1.0
<i>Trial wavefunction</i>		
--trial-state	String	required
--kappa	Float64	1.0
--jastrow-potentials	String	""
--n-max	Int64	N
<i>Stochastic reconfiguration</i>		
--eta	Float64	0.01
--seed	Int64	1234
<i>Optimization MC</i>		
--no-optimize	flag	false
--max-iters	Int64	100
--opt-num-walkers	Int64	100
--opt-num-MC-steps	Int64	5000
--opt-num-equil-steps	Int64	1000
--opt-block-size	Int64	500
<i>Final MC</i>		
--final-num-walkers	Int64	100
--final-num-MC-steps	Int64	5000
--final-num-equil-steps	Int64	1000
--final-block-size	Int64	500
<i>Output and miscellaneous</i>		
--output-dir	String	"data/VMC"
--save-history	flag	false
--skip-timing	flag	false
-h, --help	flag	—

4 Observables

In exact diagonalization, we must compute the full energy expectation value:

$$E = \langle H \rangle = \frac{\langle \psi_T | H | \psi_T \rangle}{\langle \psi_T | \psi_T \rangle}, \quad (2)$$

which is a sum over the entire Hilbert space for our system. Instead of brute force calculating the full expectation value, we can manipulate this expression to obtain a Monte Carlo-friendly form. The manipulation goes as follows:

$$\begin{aligned}
E\langle\psi_T|\psi_T\rangle &= \langle\psi_T|H|\psi_T\rangle \\
&= \sum_n \langle\psi_T|n\rangle\langle n|H|\psi_T\rangle \\
&= \sum_n (\langle\psi_T|n\rangle\langle n|H|\psi_T\rangle) \left(\frac{\langle n|\psi_T\rangle}{\langle n|\psi_T\rangle} \right) \\
&= \sum_n \frac{\langle n|H|\psi_T\rangle}{\langle n|\psi_T\rangle} |\langle n|\psi_T\rangle|^2,
\end{aligned}$$

where we have defined a quantity named the local energy as

$$E_{\text{loc}}(n) = \frac{\langle n|H|\psi_T\rangle}{\langle n|\psi_T\rangle} \quad (3)$$

and the probability as

$$p(n) = |\langle n|\psi_T\rangle|^2. \quad (4)$$

Assuming normalization, the Monte Carlo-friendly total energy expectation value is then

$$E = \sum_n p(n) E_{\text{loc}}(n), \quad (5)$$

where we have assumed normalization of the trial wavefunction: i.e., $\langle\psi_T|\psi_T\rangle = 1$. We can further expand the local energy in order to calculate it for each time we sample in the Monte Carlo simulation:

$$E_{\text{loc}}(\vec{n}) = \frac{\langle\vec{n}|H|\psi_T\rangle}{\langle\vec{n}|\psi_T\rangle} = \sum_{\vec{n}'} \frac{\langle\vec{n}|H|\vec{n}'\rangle\psi_T(\vec{n}')}{\psi_T(\vec{n})}, \quad (6)$$

which only sums over the system configurations $\vec{n}'$ that are connected to $\vec{n}$ by the Hamiltonian ($\langle\vec{n}|H|\vec{n}'\rangle$) weighted by $\psi_T(\vec{n}')/\psi_T(\vec{n})$.

5 Trial Wavefunction

Before detailing the two trial wavefunctions implemented in this code, we must define the arbitrary Fock state that mathematically describes any system configuration. For any given system configuration of linear size L , the Fock state is

$$|\vec{n}\rangle = |n_1, n_2, \dots, n_L\rangle = \prod_{i=1}^L \frac{(b^\dagger)^{n_i}}{\sqrt{n_i!}} |0\rangle. \quad (7)$$

Additionally, we can define the diagonal observables, which are independent of the trial wavefunction:

5.1 Local Potential Energy

The local potential energy is calculated as follows:

$$V_{\text{loc}}^i = \frac{\langle n|V|\psi_T\rangle}{\langle n|\psi_T\rangle} = \sum_{n'} \langle n| \frac{U}{2} n_i(n_i - 1) |n'\rangle \frac{\langle n'|\psi_T\rangle}{\langle n|\psi_T\rangle} \quad (8)$$

$$V_{\text{loc}}^i = \frac{U}{2} n_i(n_i - 1). \quad (9)$$

5.2 Local Density-Density Correlation Function

The local density-density correlation function is calculated as follows:

$$C_{\text{loc}}^{i,j} = \frac{\langle n|n_i n_j|\psi_T\rangle}{\langle n|\psi_T\rangle} = \sum_{n'} \langle n|n_i n_j|n'\rangle \frac{\langle n'|\psi_T\rangle}{\langle n|\psi_T\rangle} \quad (10)$$

$$C_{\text{loc}}^{i,j} = n_i n_j. \quad (11)$$

5.3 Gutzwiller

The Gutzwiller trial wavefunction is defined as

$$|\psi_G\rangle = \prod_j \left(\sum_{n_j=0}^{n_{\text{max}}} f_{n_j} \frac{(b_j^\dagger)^{n_j}}{\sqrt{n_j!}} \right) |0\rangle_j, \quad (12)$$

where the variational weights are parameterized as

$$f_{n_j} = \mathcal{N} e^{-\kappa n_j^2}, \quad (13)$$

with $\mathcal{N}$ a normalization constant and $\kappa > 0$ a single variational parameter. One thing you should notice about the Gutzwiller trial wavefunction is that it is a product state of separate site states: i.e., each site is independent of all others. Therefore, no correlation effects are taken into account. This can be seen as the Gutzwiller approximates the ground state energy poorly at higher interaction (U/t) values.

5.3.1 Gutzwiller Acceptance Ratio

We want to calculate the probability of going from state n to state n' :

$$P(n \rightarrow n') = \min \left(1, \frac{|\psi_G(n')|^2}{|\psi_G(n)|^2} \right). \quad (14)$$

Let's now work out this ratio. We begin by using the relation we found earlier for $\langle n|\psi_G(n)\rangle$:

$$\frac{|\psi_G(n')|^2}{|\psi_G(n)|^2} = \frac{|\prod_i^L f_{n'_i}|^2}{|\prod_j^L f_{n_j}|^2}, \quad (15)$$

where n and n' differ only by $n'_i = n_i + 1$ and $n'_{i+1} = n_{i+1} - 1$. Therefore, we find

$$\frac{|\psi_G(n')|^2}{|\psi_G(n)|^2} = \frac{f_{n_{i+1}-1} f_{n_i+1}}{|f_n|^2}. \quad (16)$$

5.3.2 Gutzwiller Local Kinetic Energy

Since the kinetic energy term is more complicated, below is a more in-depth calculation for a single term of a nearest-neighbor pair:

$$T_{\text{loc}}^i = \frac{\langle n| -tb_i^\dagger b_{i+1} - tb_{i+1}^\dagger b_i |\psi_G\rangle}{\langle n|\psi_G\rangle} \quad (17)$$

$$= -t \sum_{n'} \langle n| b_i^\dagger b_{i+1} + b_{i+1}^\dagger b_i |n'\rangle \frac{\psi_G(n')}{\psi_G(n)} \quad (18)$$

$$= -t \sum_{n'} \left[\sqrt{n'_{i+1}(n'_i + 1)} \langle n| n'_i + 1, n'_{i+1} - 1 \rangle \right] \frac{\psi_G(n')}{\psi_G(n)} \quad (19)$$

$$+ \left[\sqrt{n'_i(n'_{i+1} + 1)} \langle n| n'_i - 1, n'_{i+1} + 1 \rangle \right] \frac{\psi_G(n')}{\psi_G(n)}.$$

From here, it is obvious that only surviving terms will be when $n'_{i+1} = n_{i+1} + 1$ and $n'_i = n_i - 1$ for the first term and $n'_{i+1} = n_{i+1} - 1$ and $n'_i = n_i + 1$ for the second term. Now, we will derive the simple form of the Gutzwiller projection onto our Fock state:

$$\langle n | \psi_G \rangle = \prod_i^L \langle n_j | \left(\sum_{m_j=0}^{n_{\max}} \frac{f_{m_j} (b_j^\dagger)^{m_j}}{\sqrt{m_j!}} |0_j\rangle \right) \quad (20)$$

$$= \prod_i^L \langle n_j | \left(\sum_{m_j=0}^{n_{\max}} f_{m_j} |m_j\rangle \right), \quad (21)$$

since $(b_j^\dagger)^m |0\rangle = \sqrt{m!} |m\rangle$. We, therefore, find that

$$\langle n | \psi_G \rangle = \prod_i^L f_{n_i}. \quad (22)$$

Using this relation, the local kinetic energy for a single site is then

$$T_{\text{loc}}^i = -t \sqrt{n_{i+1}(n_i + 1)} \frac{f_{n_{i+1}+1} f_{n_{i+1}-1}}{f_{n_i} f_{n_{i+1}}} + \sqrt{n_i(n_{i+1} + 1)} \frac{f_{n_{i+1}+1} f_{n_i-1}}{f_{n_{i+1}} f_{n_i}}. \quad (23)$$

5.3.3 Gutzwiller Local Potential Energy

Since the potential term in the Hamiltonian is diagonal in the occupation basis, we can simply write is as

$$V_{\text{loc}}^i = \frac{U}{2} n_i (n_i - 1). \quad (24)$$

5.4 Jastrow

The general form of the Jastrow trial wavefunction is

$$|\psi_J\rangle = \exp \left(-\frac{1}{2} \sum_{i,j} \nu_{i,j} n_i n_j \right) |\Phi_0\rangle, \quad (25)$$

where

$$|\Phi_0\rangle = (b_{k=0}^\dagger)^N |0\rangle. \quad (26)$$

Since our system is translationally invariant, we choose to write the Jastrow potentials as follows,

$$\nu_{i,j} = \nu(r) = \nu(|R_i - R_j|), \quad (27)$$

so they only depend on the distance between sites. This results in $\text{FloorDivision}(L/2)$ unique parameters, which we need to optimize.

5.4.1 Acceptance Ratio

We want to calculate the probability of going from state n to state n' :

$$P(n \rightarrow n') = \min \left(1, \frac{|\psi_J(n')|^2}{|\psi_J(n)|^2} \right). \quad (28)$$

We begin by performing a couple preliminary steps for later use in the derivation. The first is to Fourier transform the $k = 0$ creation operator in Φ_0 as follows:

$$b_k^\dagger = \left(\frac{1}{L} \right)^{N/2} \sum_i b_i^\dagger e^{i\vec{k} \cdot \vec{r}} \Rightarrow b_{k=0}^\dagger = \frac{1}{\sqrt{L}} \sum_i b_i^\dagger. \quad (29)$$

The second is to derive a relation from our basis vector definition. See equation 7:

$$\prod_{i=1}^L (b_i^\dagger)^{n_i} |0\rangle = \prod_{i=1}^L \sqrt{n_i!} |n\rangle. \quad (30)$$

We now want to start our derivation by inserting the Fourier transform we derived earlier for $k = 0$ and expand the Φ_0 piece using the multinomial theorem described by

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{n_1, n_2, \dots, n_k \geq 0} \frac{n!}{n_1! n_2! \dots n_k!} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}, \quad (31)$$

where $n_1 + n_2 + \dots + n_k = n$. We can, therefore, write

$$(b_{k=0}^\dagger)^N = \left(\frac{1}{L}\right)^{N/2} (b_1^\dagger + b_2^\dagger + \dots + b_k^\dagger)^N = \left(\frac{1}{L}\right)^{N/2} \sum_{n_1, n_2, \dots, n_k \geq 0} \frac{N!}{n_1! n_2! \dots n_k!} (b_1^\dagger)^{n_1} (x_2^\dagger)^{n_2} \dots (x_k^\dagger)^{n_k} \quad (32)$$

Simplifying the notation, we obtain

$$|\Phi_0\rangle = \left(\frac{1}{L}\right)^{N/2} \sum_{\substack{n_1, n_2, \dots, n_k \geq 0 \\ n_1 + n_2 + \dots + n_k = N}} \frac{N!}{\prod_{j=1}^L n_j!} \prod_{j=1}^L (b_j^\dagger)^{n_j} |0\rangle \quad (33)$$

We will now make use of equation 30 we found earlier by plugging it into the above equation:

$$|\Phi_0\rangle = \left(\frac{1}{L}\right)^{N/2} \sum_{\substack{n_1, n_2, \dots, n_k \geq 0 \\ n_1 + n_2 + \dots + n_k = N}} \frac{N!}{\prod_{j=1}^L n_j!} \prod_{j=1}^L \sqrt{n_j!} |n\rangle = \left(\frac{1}{L}\right)^{N/2} \sum_{\substack{n_1, n_2, \dots, n_k \geq 0 \\ n_1 + n_2 + \dots + n_k = N}} \frac{N!}{\prod_{j=1}^L \sqrt{n_j!}} |n\rangle \quad (34)$$

We now want to project this state into a specific basis configuration for wavefunction calculation:

$$\Phi_0(n) = \langle m | \left[|\Phi_0\rangle = \left(\frac{1}{L}\right)^{N/2} \sum_{\substack{n_1, n_2, \dots, n_k \geq 0 \\ n_1 + n_2 + \dots + n_k = N}} \frac{N!}{\prod_{j=1}^L \sqrt{n_j!}} \right] |n\rangle \quad (35)$$

$$= \left(\frac{1}{L}\right)^{N/2} \sum_{\substack{n_1, n_2, \dots, n_k \geq 0 \\ n_1 + n_2 + \dots + n_k = N}} \frac{N!}{\prod_{j=1}^L \sqrt{n_j!}} \delta_{n,m} = \left(\frac{1}{L}\right)^{N/2} \frac{N!}{\prod_{j=1}^L \sqrt{n_j!}} \quad (36)$$

Now that we have a workable form of $\Phi_0(n)$, we want to look at the ratio for a nearest neighbor hop:

$$\frac{\Phi_0(n')}{\Phi_0(n)} = \frac{\frac{1}{L^{N/2}} \frac{N!}{\prod_j^L \sqrt{n_j!}}}{\frac{1}{L^{N/2}} \frac{N!}{\prod_j^L \sqrt{n_j!}}} = \prod_j^L \left(\frac{n_j!}{n_j'!} \right)^{1/2} \quad (37)$$

Since the hops are nearest neighbor, $n'_{\text{from}} = n_{\text{from}} - 1$ and $n'_{\text{to}} = n_{\text{to}} + 1$. After plugging in and simplifying factorials, we obtain

$$\frac{\Phi_0(n')}{\Phi_0(n)} = \sqrt{\frac{n_{\text{from}}}{n_{\text{to}} + 1}} \quad (38)$$

For a 1D ring of length L , the distance between sites is

$$r = (j - i) \bmod (L), \quad (39)$$

because we are working with a periodic lattice $\Rightarrow \nu(r) = \nu(L - r)$. Therefore, we can write

$$\sum_{i,j} \nu_{i,j} n_i n_j = \sum_{r=1}^{L-1} \nu(r) \sum_i^L n_i n_{i+r}. \quad (40)$$

However, it should be noticed that the second sum will double count differently for odd and even system size L . We will begin to remedy this by summing over unique distances $r = 1, 2, \dots, \frac{L}{2}$ and treating the odd and even L values separately as follows. For odd L , there is no special point where we double count, so we can write

$$- \sum_{r=1}^{\frac{L-2}{2}} \nu(r) \sum_i^L n_i n_{i+r}. \quad (41)$$

For even L , $r = \frac{L}{2}$ satisfies $\frac{L}{2} = L - \frac{L}{2}$, so we double count $\nu(\frac{L}{2})$. We can then write the even contribution as

$$- \sum_{r=1}^{\frac{L}{2}-1} \nu(r) \sum_i^L n_i n_{i+r} - \frac{1}{2} \nu\left(\frac{L}{2}\right) \sum_i^L n_i n_{i+\frac{L}{2}}. \quad (42)$$

We can combine the even and odd contributions with the following function definitions:

$$w(r) = \begin{cases} \nu(r), & r < \frac{L}{2}, \\ \frac{1}{2} \nu\left(\frac{L}{2}\right), & r = \frac{L}{2}, \end{cases} \quad (43)$$

and

$$S_r(n) = \sum_{i=1}^L n_i n_{i+r}. \quad (44)$$

Therefore, we can write

$$\log \left[\exp \left(-\frac{1}{2} \sum_{i,j} \nu_{i,j} n_i n_j \right) \right] = -\frac{1}{2} \sum_{i=1}^{\lfloor L/2 \rfloor} w_r S_r(n), \quad (45)$$

where $\lfloor L/2 \rfloor$ represents floor division. Note that we take the log for both ease of derivation and numerical stability in code. We can now work with the entire acceptance ratio:

$$\log \left(\frac{|\psi_J(n')|^2}{|\psi_J(n)|^2} \right) = 2 \log \left[\frac{\exp \left(-\frac{1}{2} \sum_{r=1}^{\lfloor L/2 \rfloor} w_r S_r(n') \right) \Phi_0(n')}{\exp \left(-\frac{1}{2} \sum_{r=1}^{\lfloor L/2 \rfloor} w_r S_r(n) \right) \Phi_0(n)} \right] \quad (46)$$

$$= 2 \left[-\frac{1}{2} \sum_{r=1}^{\lfloor L/2 \rfloor} w_r S_r(n') + \frac{1}{2} \sum_{r=1}^{\lfloor L/2 \rfloor} w_r S_r(n) + 2 \log \left(\sqrt{\frac{n_{\text{from}}}{n_{\text{to}} + 1}} \right) \right] \quad (47)$$

$$\log \left(\frac{|\psi_J(n')|^2}{|\psi_J(n)|^2} \right) = \left[\sum_{r=1}^{\lfloor L/2 \rfloor} w_r S_r(n) - \sum_{r=1}^{\lfloor L/2 \rfloor} w_r S_r(n') \right] + \log(n_{\text{from}}) - \log(n_{\text{to}} + 1). \quad (48)$$

Note that in the sum subtraction we only need the difference between sites where particle number changed via hopping move.

5.4.2 Jastrow Local Kinetic Energy

Similar to our energy calculations from the Gutzwiller trial wavefunction, we begin with the following form:

$$T_{\text{loc}}^i = \frac{\langle n | T | \psi_J \rangle}{\langle n | \psi_J \rangle} = \sum_{n'} \langle n | T | n' \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle}, \quad (49)$$

where the ratio on the right-hand side is simply the square root of the acceptance ratio we compute earlier. We can continue computing this explicitly to obtain the prefactors:

$$T_{\text{loc}}^i = -t \sum_{n'} \langle n | b_i^\dagger b_{i+1} + b_{i+1}^\dagger b_i | n' \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle} \quad (50)$$

$$\begin{aligned} &= -t \sum_{n'} \sqrt{n'_{i+1}(n'_i + 1)} \langle n | n'_{i+1} - 1, n'_i + 1 \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle} \\ &\quad - t \sum_{n'} \sqrt{n'_i(n'_{i+1} + 1)} \langle n | n'_i - 1, n'_{i+1} + 1 \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle}. \end{aligned} \quad (51)$$

From here, it is obvious that only terms where $n'_i + 1 = n_i$, $n'_{i+1} - 1 = n_{i+1}$ and $n'_i - 1$, $n'_{i+1} + 1 = n_{i+1}$ will survive for the left and right sums respectively. Our final expression is then

$$T_{\text{loc}}^i = -t \sqrt{(n_{i+1} + 1)n_i} \frac{\langle n_{i+1} + 1, n_i - 1 | \psi_J \rangle}{\langle n | \psi_J \rangle} - t \sqrt{(n_i + 1)n_{i+1}} \frac{\langle n_i + 1, n_{i+1} - 1 | \psi_J \rangle}{\langle n | \psi_J \rangle}. \quad (52)$$

6 VMC Algorithm

6.1 Markov Chain

A Markov chain is a sequence of random variables $X_1, X_2, \dots, X_n$ where the probability of being in a state X_{n+1} depends only on the previous state X_n . The Markov chain is defined by a transition matrix P where each element P_{ij} is the probability of transitioning from state i to state j . The transition matrix must satisfy the following properties:

1. $P_{ij} \geq 0$ for all i, j
2. $\sum_j P_{ij} = 1$ for all i

6.2 Metropolis–Hastings sampling

6.2.1 Detailed Balance

These Markov chains are built from two pieces: a proposal probability $g(n \rightarrow n')$ and an acceptance probability $A(n \rightarrow n')$. The proposal probability is the probability of proposing a new configuration n' given the current configuration n . The acceptance probability is the probability of accepting the proposed configuration. The product of these two pieces gives the transition probability from n to n' :

$$P(n \rightarrow n') = g(n \rightarrow n') A(n \rightarrow n'). \quad (53)$$

The transition probability must satisfy the detailed balance condition:

$$p(n) P(n \rightarrow n') = p(n') P(n' \rightarrow n), \quad (54)$$

where $p(n)$ is the probability of being in configuration n described by our sampling distribution. Since there is an infinite set of solutions to this equation, we pin the value of the backward move ($n' \rightarrow n$) to 1 and rearrange this equation to find the forward move ($n \rightarrow n'$) acceptance probability:

$$A(n \rightarrow n') = \frac{p(n') g(n' \rightarrow n)}{p(n) g(n \rightarrow n')}. \quad (55)$$

Since our proposals are symmetric, $g(n \rightarrow n') = g(n' \rightarrow n)$, we can simplify the acceptance ratio to

$$A(n \rightarrow n') = \frac{p(n')}{p(n)}. \quad (56)$$

In the case of variational Monte Carlo, we have $p(n) = |\psi_T(n)|^2$, so the acceptance ratio is

$$A(n \rightarrow n') = \frac{|\psi_T(n')|^2}{|\psi_T(n)|^2}. \quad (57)$$

6.3 Parameter optimization

We optimize the variational parameters of our trial wavefunction by performing gradient descent with stochastic reconfiguration on the energy expectation value. The gradient of the energy with respect to a variational parameter α is given by

$$f_k = -2 \left\langle \left(\hat{H} - E_\alpha \right) \hat{O}_k \right\rangle, \quad (58)$$

where $\hat{O}_k = \frac{\partial}{\partial \alpha_k} \log \psi_T$ and $E_\alpha = \langle \hat{H} \rangle$. The stochastic reconfiguration method is then used to update the variational parameters as follows:

$$\alpha_k^{(n+1)} = \alpha_k^{(n)} - \eta \sum_j S_{kj}^{-1} f_j, \quad (59)$$

where η is the learning rate and $S_{kj} = \langle \hat{O}_k \hat{O}_j \rangle - \langle \hat{O}_k \rangle \langle \hat{O}_j \rangle$ is the covariance matrix of the operators $\hat{O}_k$ and $\hat{O}_j$.

7 Code Structure

File	Responsibility
<code>lattice.jl</code>	Defines lattice and adjacency matrix
<code>MC_integration.jl</code>	Monte Carlo simulation
<code>moves.jl</code>	Metropolis–Hastings acceptance ratio
<code>measurements.jl</code>	Local energy, energy gradient, and $\Delta \log \psi$ calculations
<code>gradient_descent.jl</code>	Gradient descent with stochastic reconfiguration
<code>system.jl</code>	Stores all system parameters and lattice
<code>wavefunction.jl</code>	Stores wavefunction parameters
<code>VMCBoseHubbard.jl</code>	Module including all imports and exports
<code>main.jl</code>	Parse terminal input and run VMC

8 Benchmarks and Validation

Describe any checks against exact diagonalization, DMRG, or published results. Include convergence plots (energy vs. Monte Carlo steps, energy vs. variational parameter) as figures.

We have benchmarked our Gutzwiller and Jastrow trial wavefunctions by comparing ground state energy VMC results with ground state energy results from an exact diagonalization code for system sizes up to $L = 12$, $N = 12$. In addition, we confirmed the optimal Gutzwiller parameter for several interaction strengths by digitizing Werner Krauth’s Gutzwiller results from his paper [INCLUDE REFERENCE]. Below are the benchmarking graphs described above:

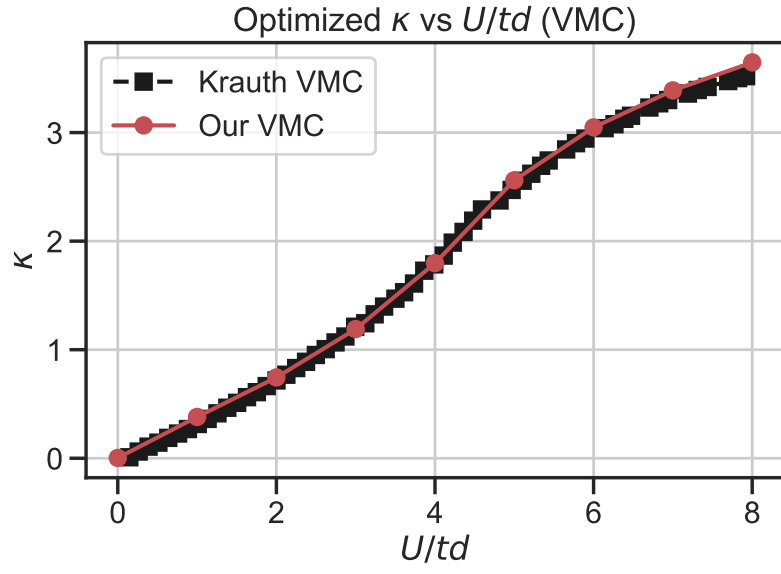


Figure 1:

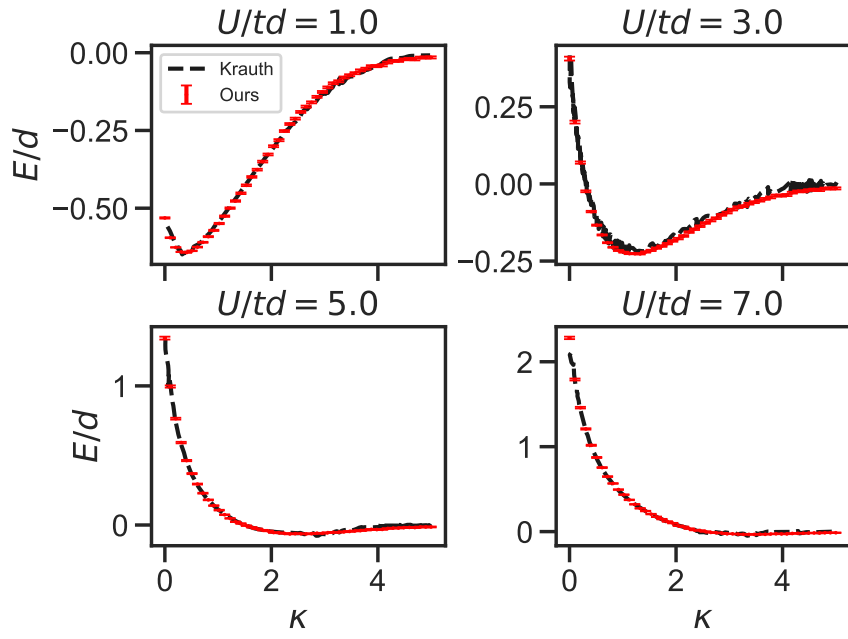


Figure 2:

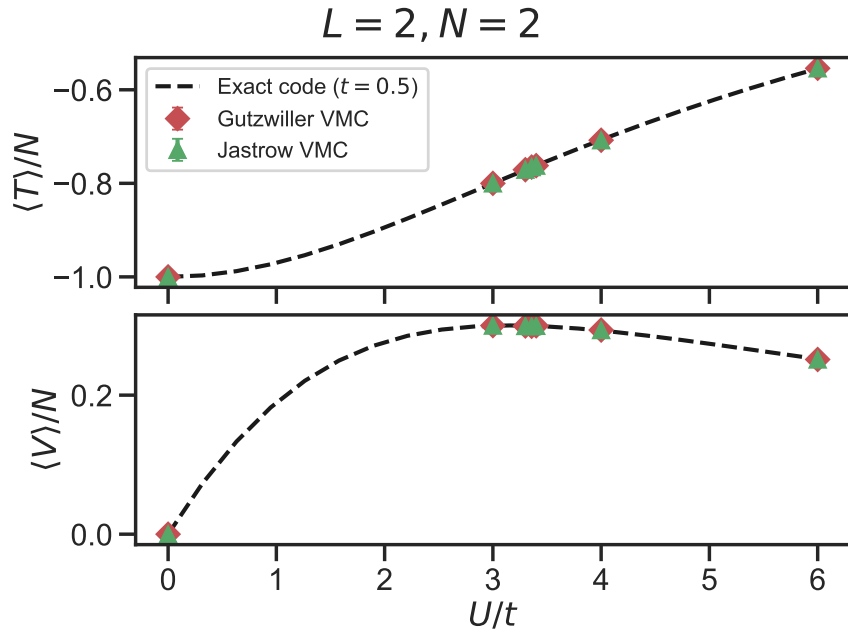


Figure 3:

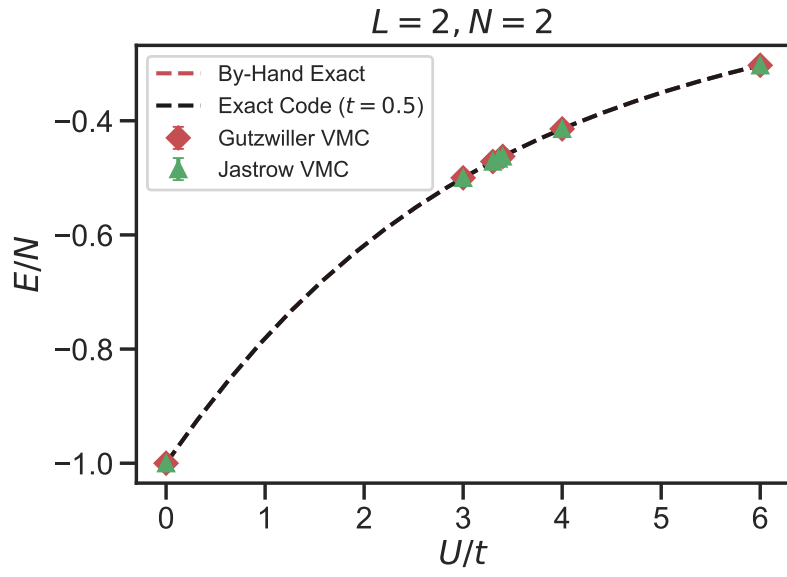


Figure 4:

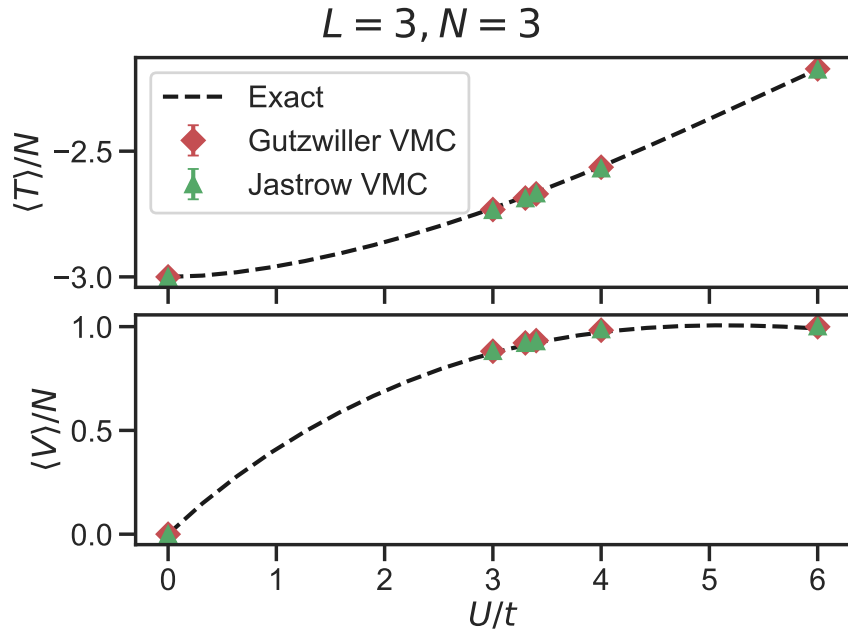


Figure 5:

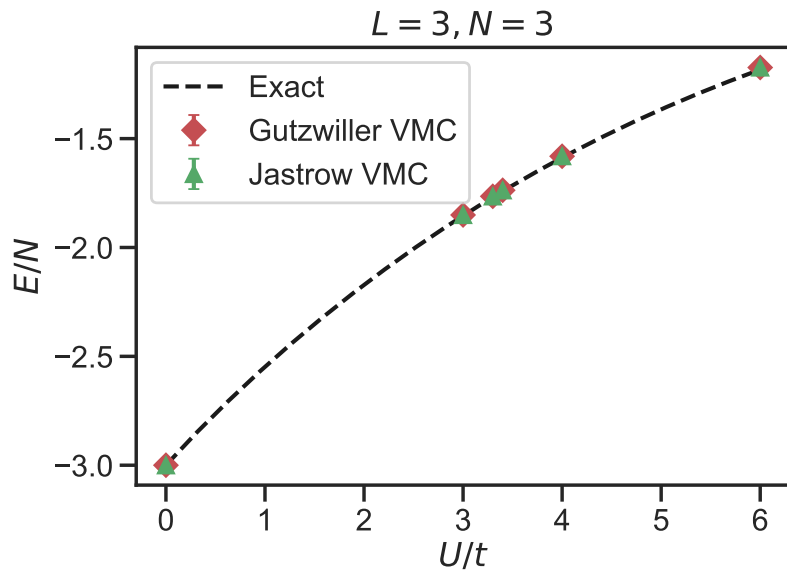


Figure 6:

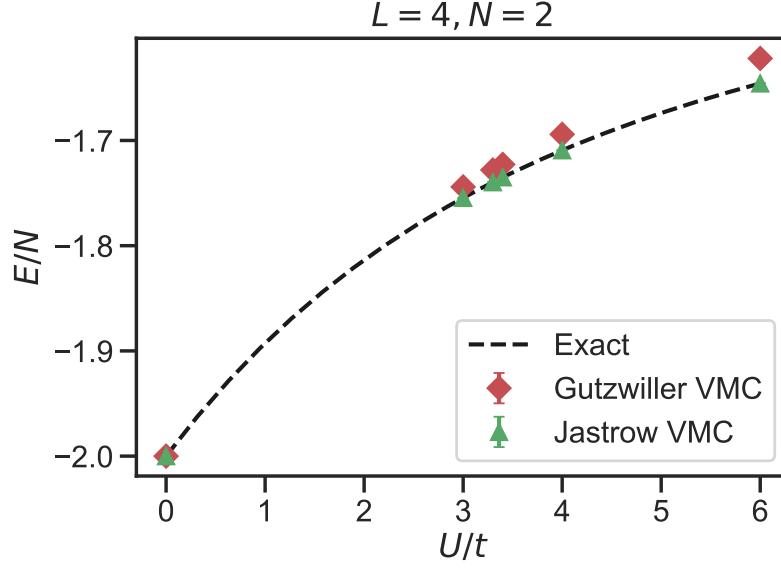


Figure 7:

9 Known Limitations and Future Work

- The code is currently limited to only one-dimensional systems
- The Gutzwiller trial wavefunction does not take correlations between sites into account, causing it to fail in the high interaction (Mott insulating) regime

References